

Mr RANDATH

↑ AFFIX SEAL HERE ↑

CANDIDATE – PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

Signature

TEST CODE **01238010**

MAY/JUNE 2007

FORM TP2007116

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

PHYSICS

Paper 01 – General Proficiency

75 minutes

28 MAY 2007 (a.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
3. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) newton
- (B) metre
- (C) kilogram
- (D) second

Sample Answer

(A) ☒ (C) (D)

The best answer to this item is “metre” so answer space (B) has been shaded.

4. If you want to change your answer, be sure to erase your old answer completely and fill in your new choice.
5. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. You can come back to the harder item later. Your score will be the total number of correct answers.
6. Figures are not necessarily drawn to scale.
7. You may do any rough work in this booklet.
8. The use of non-programmable calculators is allowed.
9. This test consists of 60 items. You will have 75 minutes to answer them.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following is an SI base unit?

- (A) Ampere
- (B) Volt
- (C) Ohm
- (D) Coulomb

2. To measure the external diameter of a measuring cylinder most accurately one should use a

- (A) tape measure
- (B) metre rule
- (C) length of string
- (D) pair of vernier callipers

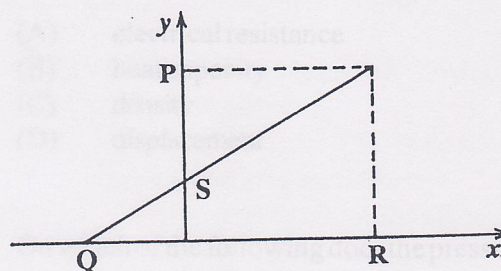
3. When used in front of a unit the prefix 'mega' means

- (A) 10^6
- (B) 10^3
- (C) 10^{-3}
- (D) 10^{-6}

4. Which of the following is true about the period of a simple pendulum?

- (A) It increases with the length of the pendulum
- (B) It is dependent on the mass of the bob
- (C) It changes with the volume of the bob
- (D) It decreases with decreasing amplitude

Item 5 refers to the diagram below.



5. When $x = 0$, the value of y is

- (A) Q
- (B) S
- (C) P
- (D) P/R

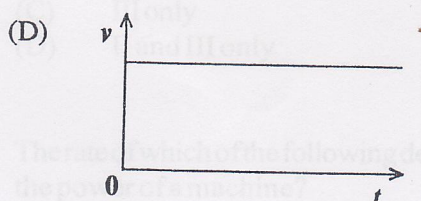
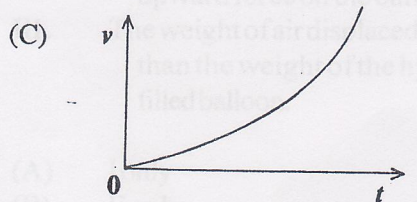
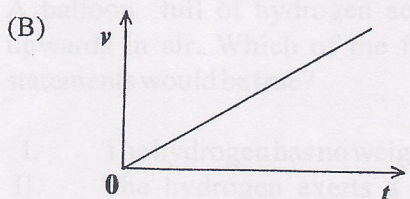
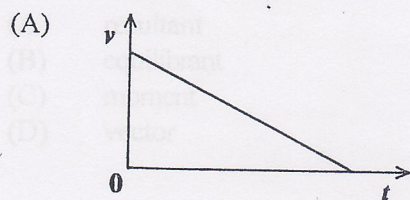
6. Which of the following is a fundamental quantity?

- (A) Density
- (B) Mass
- (C) Relative density
- (D) Volume

7. Which of the following is NOT a derived quantity?

- (A) Energy
- (B) Force
- (C) Charge
- (D) Current

8. A coconut falls from a tree. Which of the velocity/time graphs below BEST represents its motion, if air resistance is neglected?



9. A bus, with luggage loaded on top, is more likely to tip over when rounding a corner than the same bus without the luggage. The reason for this is that the luggage

- (A) increases the weight of the bus
- (B) raises the centre of gravity of the bus
- (C) lowers the centre of gravity of the bus
- (D) increases the momentum of the bus

10. An example of a vector quantity is

- (A) electrical resistance
- (B) heat capacity
- (C) density
- (D) displacement

11. On which of the following does the pressure at a point in a liquid depend?

- I. Density of the liquid
- II. Depth from the surface
- III. Area of the cross-section of the container

- (A) I only
- (B) III only
- (C) I and II only
- (D) I, II and III

12. Which of the following is the unit of momentum?

- (A) m s^{-1}
- (B) kg m s^{-1}
- (C) kg m s^{-2}
- (D) N s

13. To which of the following types of energy does a car, travelling on a level road, convert its kinetic energy when it slows down and stops, using its brakes?

- I. Thermal
- II. Potential
- III. Gravitational

- (A) I only
- (B) II only
- (C) I and II only
- (D) I and III only

14. A single force, which can replace and have the same effect as two or more forces acting together, is known as the

(A) resultant
(B) equilibrant
(C) moment
(D) vector

15. A balloon full of hydrogen accelerates upwards in air. Which of the following statements would be true?

I. The hydrogen has no weight.
II. The hydrogen exerts a resultant upward force on the balloon.
III. The weight of air displaced is greater than the weight of the hydrogen-filled balloon.

(A) I only
(B) II only
(C) III only
(D) II and III only

16. The rate of which of the following determines the power of a machine?

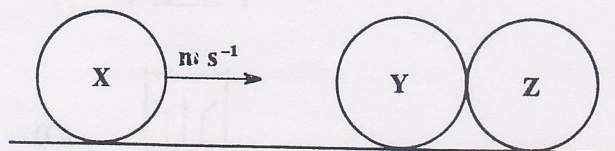
I. Doing work
II. Converting energy
III. Changing temperature

(A) I only
(B) III only
(C) I and II only
(D) I, II and III

17. The rate of which of the following is NOT true about the moment of a force?

(A) It is measured in units of newtonmetres.
(B) It is the amount of force needed to keep a body turning.
(C) It is the turning effect that a force has when it acts on a body.
(D) It is the product of a force and the perpendicular distance of its line of action from a point.

Item 18 refers to the following diagram



18. X, Y and Z are three balls of equal mass whose centres lie in a straight line. Y and Z are touching one another. X, travelling with a velocity of $v \text{ m s}^{-1}$, hits on Y. Assuming that the spheres are perfectly elastic, the resultant motion after impact will be that

(A) X, Y and Z all move on together, with velocity $v \text{ m s}^{-1}$
(B) X and Y both remain stationary while Z moves on with a velocity of $v \text{ m s}^{-1}$
(C) X, Y and Z remain stationary
(D) X remains stationary while Y and Z move on together, with velocity of $v \text{ m s}^{-1}$

19. Which of the following BEST explains Brownian motion of smoke particles in air?

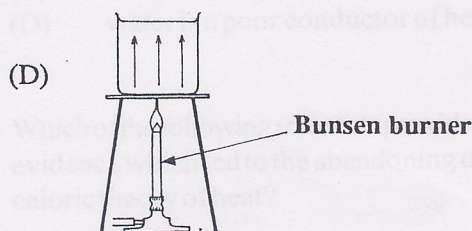
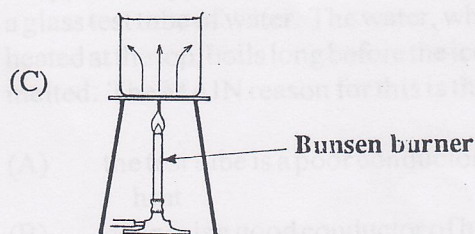
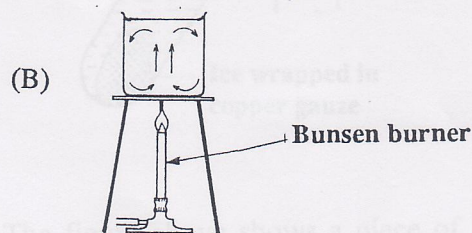
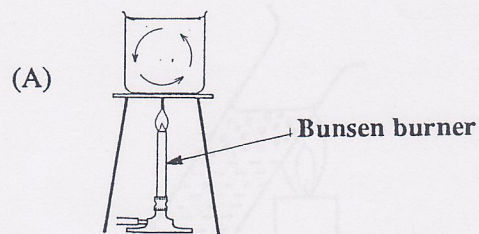
- (A) The intense beam of light causes the smoke particles to vibrate.
- (B) Smoke particles are colliding with each other.
- (C) Convection currents of air cause the vibration of smoke particles.
- (D) The air molecules are moving and colliding with the smoke particles.

20. In the pressure law, which of the following would be true?

- I. Ratio of pressure to Kelvin temperature is constant.
- II. Volume is constant.
- III. Pressure is constant.

- (A) I only
- (B) III only
- (C) I and II only
- (D) I, II and III

21. Which of the following diagrams BEST illustrates convection currents in a liquid?



22. In a thermocouple thermometer, which thermometric property is used?

- (A) Variation of electrical resistance
- (B) Expansion of metals
- (C) Emission of electrons from a hot metal surface
- (D) Production of a current at the junction of unlike metals

23. A gas is much easier to compress than a solid or liquid because the gas molecules

(A) are very numerous
(B) are arranged randomly
(C) are moving very rapidly
(D) have no forces between them

24. Which of the following would be the unit of the heat capacity of an object?

(A) J
(B) J kg^{-1}
(C) J K^{-1}
(D) $\text{J kg}^{-1} \text{K}^{-1}$

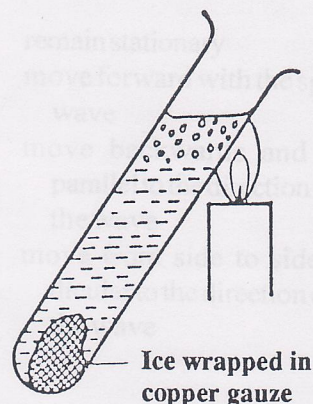
25. Which of the following is true of evaporation?

(A) It occurs throughout a liquid at no definite temperature.
(B) It occurs at the surface of a liquid at no definite temperature.
(C) It occurs at the surface of a liquid at a definite temperature.
(D) It occurs throughout a liquid at a definite temperature.

26. When the volume of fixed mass of a gas is reduced at a constant temperature, the average speed of the molecules

(A) remains the same as the temperature does not change
(B) increases as the molecules have less space in which to move
(C) increases as the molecules hit each other more often
(D) decreases as the molecules have less space in which to move

- 27.



The figure above shows a piece of ice wrapped in copper gauze and submerged in a glass test tube of water. The water, when heated at the top, boils long before the ice is melted. The MAIN reason for this is that

(A) the test tube is a poor conductor of heat
(B) copper is a good conductor of heat
(C) ice is less dense than water
(D) water is a poor conductor of heat

28. Which of the following scientists provided the evidence which led to the abandoning of the caloric theory of heat?

I. Newton
II. Boyle
III. Rumford

(A) III only
(B) II only
(C) I and III only
(D) II and III only

29. Which of the following statements would be true?

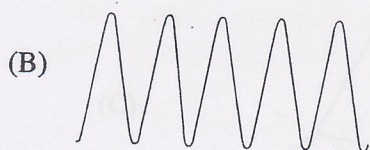
- I. Thermal radiation is an electromagnetic radiation.
- II. Blue light has a longer wave-length than infrared radiation.
- III. The velocity of X-rays is the same as the velocity of light.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

30.



The figure above shows the profile of a water wave. Using the same scale, which diagram below represents a wave TWICE the frequency and HALF the amplitude in the same tank of water?



31. In a longitudinal wave, the particles

- (A) remain stationary
- (B) move forward with the speed of the wave
- (C) move backwards and forwards parallel to the direction of travel of the wave
- (D) move from side to side, perpendicular to the direction of travel of the wave

32. Plane water waves which travel from deep water to shallow water

- (A) are diffracted
- (B) increase their speed
- (C) undergo a change in frequency
- (D) undergo a change in wave-length

33. Which of the following statements would be true?

- I. A note of frequency 512 Hz has a higher pitch than one of frequency 256 Hz.
- II. A high pitched note travels faster than a low pitched note.
- III. If the pitch of a note is increased then the periodic time of the vibrations is also increased.

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) II and III only

34. Which of the following statements about waves is true?

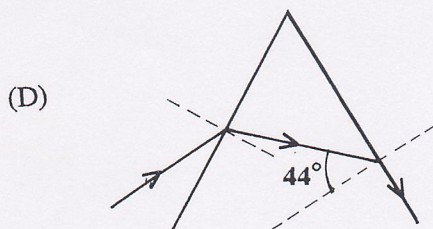
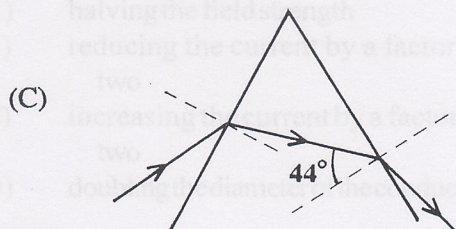
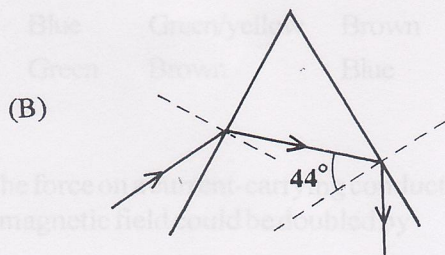
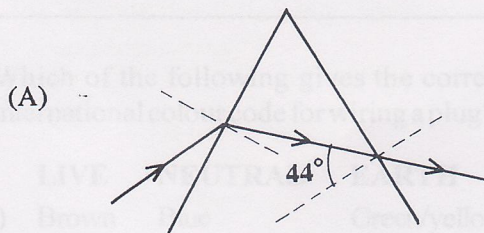
- (A) Only transverse waves undergo reflection.
- (B) Longitudinal waves do not undergo refraction, but may be reflected.
- (C) Diffraction can only take place with light waves.
- (D) All waves undergo reflection, refraction and diffraction.

35. Which of the following would be true about the image formed by a plane mirror?

- I. It is virtual.
- II. It is laterally inverted.
- III. It is magnified.

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

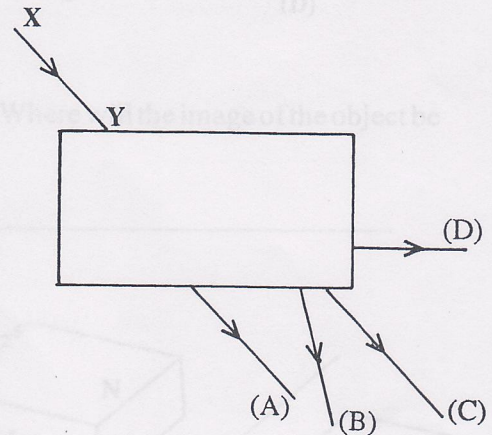
36. The critical angle of glass is 42° . Which of the following diagrams correctly shows the path of a ray of light through a triangular glass prism?



37. The image on the film of a camera is usually

- (A) inverted, diminished, real
- (B) inverted, diminished, virtual
- (C) upright, diminished, real
- (D) upright, magnified, real

38. A ray, XY, enters a rectangular glass block as shown in the diagram below. Along which direction will the ray emerge?



39. When looking into a periscope the image seen is

- (A) upside down
- (B) virtual
- (C) laterally inverted
- (D) larger than the object

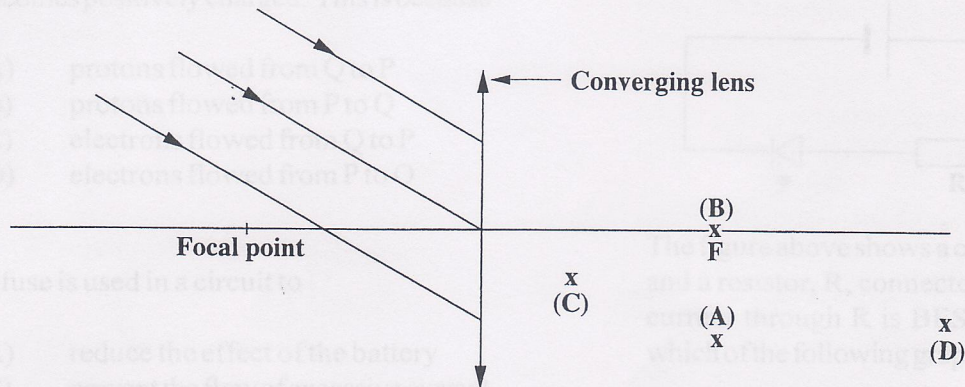
40. In the human eye, most refraction occurs at the

- (A) air/cornea boundary
- (B) aqueous humour/lens boundary
- (C) aqueous humour/cornea boundary
- (D) retina

41.

A plastic rod, P, is rubbed with cloth, Q. P becomes positively charged. This is because

- (A) protons flow from Q to P
- (B) protons flow from P to Q
- (C) electrons flow from Q to P
- (D) electrons flow from P to Q

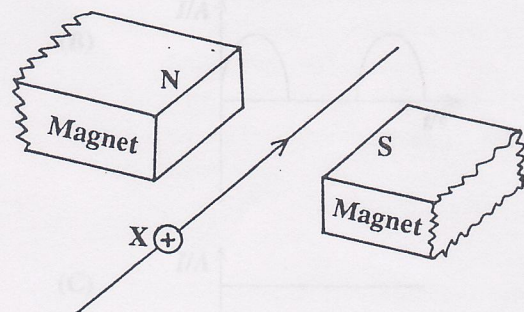


In the diagram above, the incident rays of light are parallel. Where will the image of the object be formed?

42. Which of the following gives the correct international colour code for wiring a plug?

	LIVE	NEUTRAL	EARTH
(A)	Brown	Blue	Green/yellow
(B)	Red	Blue	Green
(C)	Blue	Green/yellow	Brown
(D)	Green	Brown	Blue

44.



43. The force on a current-carrying conductor in a magnetic field could be doubled by

- (A) halving the field strength
- (B) reducing the current by a factor of two
- (C) increasing the current by a factor of two
- (D) doubling the diameter of the conductor

A positively charged particle, X, is moving at high speed in the direction indicated in the diagram above. In which direction will the magnetic field deflect the particle?

- (A) Upward
- (B) To the left
- (C) Downward
- (D) To the right

49. Which of the following is MOST suitable for the core of an electro-magnet?

- (A) Copper
- (B) Steel
- (C) Carbon
- (D) Soft iron

45. A plastic rod, P, is rubbed with cloth, Q. P becomes positively charged. This is because

- (A) protons flowed from Q to P
- (B) protons flowed from P to Q
- (C) electrons flowed from Q to P
- (D) electrons flowed from P to Q

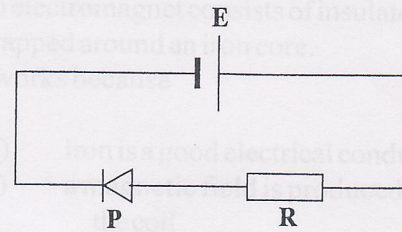
46. A fuse is used in a circuit to

- (A) reduce the effect of the battery
- (B) prevent the flow of excessive current
- (C) bridge a gap in the circuit
- (D) act as an insulator

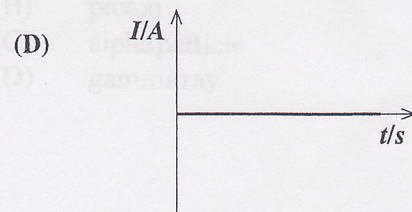
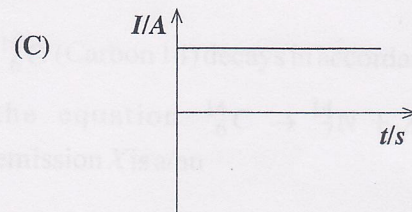
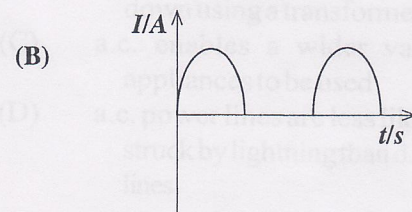
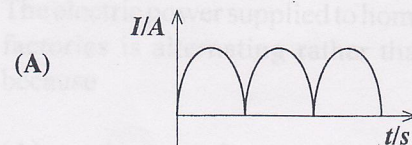
47. Which of the following circuit symbols represents a fuse?

- (A) 
- (B) 
- (C) 
- (D) 

48.



The figure above shows a cell, E, a diode, P, and a resistor, R, connected in series. The current through R is BEST illustrated by which of the following graphs?



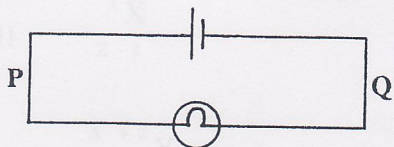
49. Which of the following is MOST suitable for the core of an electro-magnet?

- (A) Copper
- (B) Steel
- (C) Carbon
- (D) Soft iron

50. Rectification can BEST be done by using a

- (A) transformer
- (B) capacitor
- (C) transistor
- (D) diode

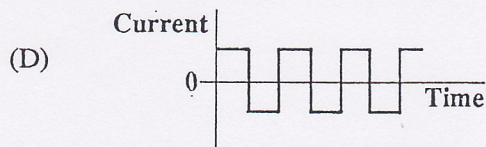
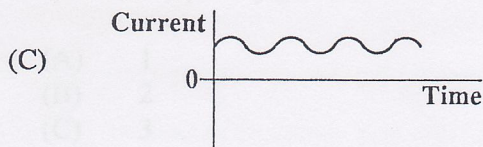
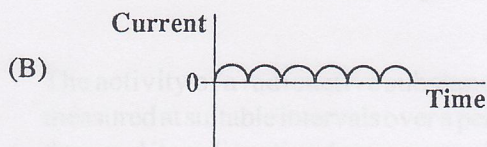
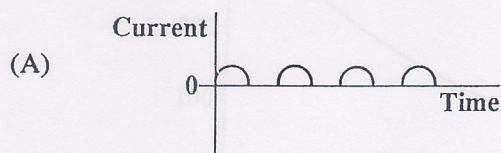
Item 51 refers to the following diagram.



51. The current in wire P is 0.9 A. The current in wire Q is

- (A) 0 A
- (B) 0.8 A
- (C) 0.9 A
- (D) 1.0 A

52. Which of the following graphs illustrates an ALTERNATING current?



53. An electromagnet consists of insulated wire wrapped around an iron core. It works because

- (A) iron is a good electrical conductor
- (B) a magnetic field is produced inside the coil
- (C) an electric field is produced inside the coil
- (D) iron is always magnetised

54. The electric power supplied to homes and to factories is alternating rather than direct because

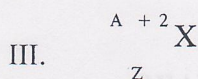
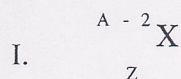
- (A) the use of a.c. reduces electrical hazards
- (B) a.c. voltages can be stepped up or down using a transformer
- (C) a.c. enables a wider variety of appliances to be used
- (D) a.c. power lines are less likely to be struck by lightning than d.c. power lines

55. $^{14}_6\text{C}$ (Carbon 14) decays in accordance with the equation $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + X$. The emission X is a/an

- (A) beta particle
- (B) proton
- (C) alpha particle
- (D) gamma ray

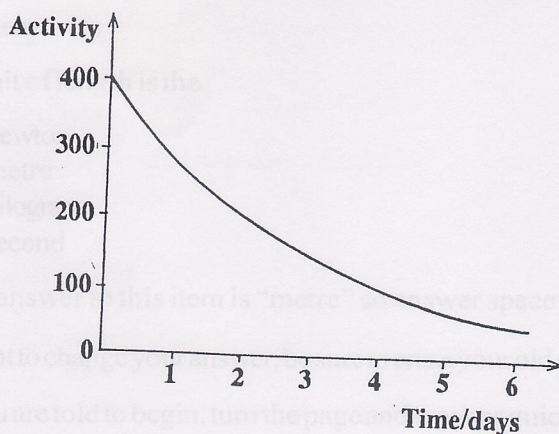
56. Which of the following would be possible symbols for an isotope of nuclide presented

by ${}^A_Z X$?



- (A) I only
(B) III only
(C) II only
(D) I and III only

57.



The activity of a radioactive substance was measured at suitable intervals over a period of days and its radioactive decay curve plotted. The half life, in days, is

- (A) 1
(B) 2
(C) 3
(D) 4

58. Radioactivity takes place in an atom as a result of

- (A) nuclear instability
(B) a chemical reaction
(C) exposure to excess heat
(D) excess stress on the atom

59. Several different types of emissions were directed at right angles to a magnetic field, and emerged undeflected. The emissions consisted of

- (A) an electron beam
(B) alpha-particles
(C) beta-particles
(D) gamma-rays

60. Which of the following statements about a proton is NOT correct?

- (A) It is a hydrogen atom minus an electron.
(B) It has the same mass as that of an electron.
(C) It has a mass about 2 000 times that of an electron.
(D) It has a charge equal in size but opposite in sign to that of an electron.